

MAP- 2202
Real Analysis - II
Tutorial - I

ASP/19/20/053

88%

(100) (i) Determine convergence or divergence of each of the following series.

(i) $\sum_{n=1}^{\infty} \sqrt{\frac{n}{2(n+1)}}$

In here $a_n = \sqrt{\frac{n}{2(n+1)}}$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sqrt{\frac{n}{2(n+1)}}$$

$$= \lim_{n \rightarrow \infty} \sqrt{\frac{n}{2n+2}}$$

$$= \lim_{n \rightarrow \infty} \sqrt{\frac{n/n}{2n/n+2/n}}$$

$$= \sqrt{\frac{1}{2}}$$

$$\lim_{n \rightarrow \infty} a_n = \sqrt{1/2}$$

since, $\lim_{n \rightarrow \infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \sqrt{\frac{n}{2(n+1)}}$ is divergent.

(ii) $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n^2}\right)$

In here, $a_n = \cos\left(\frac{1}{n^2}\right)$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \cos\left(\frac{1}{n^2}\right)$$

$$= \cos\left[\frac{1}{\infty^2}\right]$$

$$= \cos(0)$$

$$\lim_{n \rightarrow \infty} a_n = 1$$

since, $\lim_{n \rightarrow \infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n^2}\right)$ is divergent.

$$(ii) \sum_{n=1}^{\infty} \sqrt{\frac{3n^2 + 5n + 4}{4n^2 + 1}}$$

In here, $a_n = \sqrt{\frac{3n^2 + 5n + 4}{4n^2 + 1}}$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sqrt{\frac{3n^2 + 5n + 4}{4n^2 + 1}}$$

$$= \lim_{n \rightarrow \infty} \sqrt{\frac{3n^2/n^2 + 5n/n^2 + 4/n^2}{4n^2/n^2 + 1/n^2}}$$

$$= \sqrt{\frac{3}{4}}$$

$$\lim_{n \rightarrow \infty} a_n = \frac{\sqrt{3}}{2}$$

since, $\lim_{n \rightarrow \infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \sqrt{\frac{3n^2 + 5n + 4}{4n^2 + 1}}$ is divergent.

$$(i) \sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^{1/n}$$

In here, $a_n = \left(\frac{1}{n}\right)^{1/n}$

$$a_n = \left(\frac{1^{1/n}}{n^{1/n}}\right)$$

$$a_n = \frac{1}{n^{1/n}}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n^{1/n}}\right)$$

$$\lim_{n \rightarrow \infty} a_n = 1 \quad (\text{since } \lim_{n \rightarrow \infty} n^{1/n} = 1)$$

since, $\lim_{n \rightarrow \infty} a_n \neq 0$, the series $\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^{1/n}$ is divergent.

$$\textcircled{1} \sum_{n=1}^{\infty} \left[\frac{2}{(-1)^n - 3} \right]^n$$

Let $a_n = \left[\frac{2}{(-1)^n - 3} \right]^n$ for all $n \in \mathbb{N}$

$$|a_n| = \left[\frac{2}{(-1)^n - 3} \right]^n$$

$$|a_n|^{1/n} = \frac{2}{(-1)^n - 3} \text{ for all } n \in \mathbb{N}$$

$$|a_n|^{1/n} = \begin{cases} -2/2 = -1 & ; \text{ if } n \text{ is even} \\ 2/-4 = -1/2 & ; \text{ if } n \text{ is odd} \end{cases}$$

$$A = \{ |a_1|^{1/n}, |a_2|^{1/n}, |a_3|^{1/n}, \dots, |a_n|^{1/n}, \dots \}$$

$$A = \{ -1/2, -1, -1/2, \dots \}$$

$$\sup A = -1/2$$

$$\alpha = \limsup |a_n|^{1/n}$$

$$\alpha = \lim_{n \rightarrow \infty} \left[\sup_{k \geq n} |a_k|^{1/k} \right]$$

$$\alpha = \lim_{n \rightarrow \infty} -1/2$$

$$\alpha = -1/2 < 1$$

$\therefore$ By the root test, the series is absolutely convergent.
 $\therefore$ It is convergent.

② Determine the convergence or divergence of the following series.

60
① $\sum_{n=1}^{\infty} \frac{1}{n^2+25}$

In here, $a_n = \frac{1}{n^2+25}$ for all $n \in \mathbb{N}$

$$a_{n+1} = \frac{1}{(n+1)^2+25}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{1}{(n+1)^2+25} \times \frac{n^2+25}{1} \right|$$

$$= \left| \frac{n^2+25}{(n+1)^2+25} \right|$$

$$= \frac{n^2+25}{n^2+2n+1+25}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{n^2+25}{n^2+2n+26}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$L = \lim_{n \rightarrow \infty} \left(\frac{n^2+25}{n^2+2n+26} \right)$$

$$L = 1 \quad \checkmark$$

• Limit form of the ratio test cannot be applied.

then we use another test if it is not say, we use ratio test

iii) $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin\left(\frac{1}{n^2}\right)$

In here $a_n = \frac{1}{n^2} \sin\left(\frac{1}{n^2}\right)$

$a_n \leq \frac{1}{n^2}$

i.e. $\frac{1}{n^2} \sin\left(\frac{1}{n^2}\right) \leq \frac{1}{n^2}$ (for all $n \geq 1$)

$\sum_{n=1}^{\infty} \frac{1}{n^2} \sin\left(\frac{1}{n^2}\right) \leq \sum_{n=1}^{\infty} \frac{1}{n^2}$

By the comparison test, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent series, so $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin\left(\frac{1}{n^2}\right)$ is also convergent.

iv) $\sum_{n=2}^{\infty} \frac{n}{n^2+3}$

Let $a_n = \frac{n}{n^2+3}$ for all $n \in \mathbb{N}$

$\frac{n}{n^2+3} < \frac{n}{n^2} = \frac{1}{n} = b_n$

Let $b_n = 1/n$.

Now $L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$

$L = \lim_{n \rightarrow \infty} \left[\frac{\frac{n}{n^2+3}}{1/n} \right]$

$L = \lim_{n \rightarrow \infty} \left[\frac{n}{n^2+3} \times \frac{n}{1} \right]$

$L = \lim_{n \rightarrow \infty} \left[\frac{n^2}{n^2+3} \right]$

$L = \lim_{n \rightarrow \infty} \left[\frac{1}{1+3/n^2} \right]$

$L = 1 \in (0, \infty)$

Also we know that $\sum_{n=1}^{\infty} 1/n$ is divergent. Therefore by the limit form of comparison test the series $\sum_{n=2}^{\infty} \frac{n}{n^2+3}$ is divergent.

③ Use the ratio test to determine the convergence of each of the following series.

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i) $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

We apply the limit form of the ratio test to test convergence of the series.

$$a_n = \frac{n!}{n^n}$$

$$a_{n+1} = \frac{(n+1)!}{(n+1)^{n+1}}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(n+1)!}{(n+1)^{n+1}} \times \frac{n^n}{n!} \right|$$

$$= \left| \frac{(n+1) \cancel{n!} \times n^n}{(n+1)^{n+1} \cancel{n!}} \right|$$

$$= \left| \frac{(n+1) n^n}{(n+1)^{n+1}} \right|$$

$$= \left| (n+1)^{1-n-1} n^n \right|$$

$$= \left| (n+1)^{-n} n^n \right|$$

$$= \left| \frac{n^n}{(n+1)^n} \right|$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left(\frac{n}{n+1} \right)^n$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{1 + 1/n} \right)^n$$

$$= \lim_{n \rightarrow \infty} \left[\frac{1}{(1 + 1/n)^n} \right]$$

$$= \frac{1}{e} < 1$$

∴ By the limit form of the ratio test, the series a is absolutely convergent.
∴ It is convergent.

$$ii) \sum_{n=1}^{\infty} \frac{n!}{3 \cdot 5 \cdot 7 \dots (2n+1)}$$

We apply the limit form of the ratio test to test convergence of the series,

$$a_n = \frac{n!}{3 \cdot 5 \cdot 7 \dots (2n+1)} \quad \text{for all } n \in \mathbb{N}$$

$$a_{n+1} = \frac{(n+1)!}{3 \cdot 5 \cdot 7 \dots (2n+3)}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(n+1)!}{3 \cdot 5 \cdot 7 \dots (2n+3)} \times \frac{3 \cdot 5 \cdot 7 \dots (2n+1)}{n!} \right|$$

$$= \left| \frac{n! (n+1)}{(2n+3) n!} \right|$$

$$= \left| \frac{(n+1)}{(2n+3)} \right|$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{n+1}{2n+3} \right|$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{n+1}{2n+3} \right|$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{1 + 1/n}{2 + 3/n} \right|$$

$$L = 1/2 < 1$$

By the limit form of the ratio test, the series is absolutely convergent.
 $\therefore$ It is convergent.

$$(ii) \quad \frac{1^2 \cdot 2^2}{1!} + \frac{2^2 \cdot 3^2}{2!} + \frac{3^2 \cdot 4^2}{3!} + \dots$$

In here $a_n = \frac{n^2 \cdot (n+1)^2}{n!}$

$$a_{n+1} = \frac{(n+1)^2 \cdot (n+2)^2}{(n+1)!}$$

We apply the limit form of the ratio test to test convergence of the series.

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(n+1)^2 \cdot (n+2)^2}{(n+1)!} \times \frac{n!}{n^2 \cdot (n+1)^2} \right|$$

$$= \left| \frac{(n+2)^2 \cdot \cancel{n!}}{(n+1) \cdot \cancel{n!} \cdot n^2} \right|$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(n+2)^2}{n^2 (n+1)} \right|$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+2)^2}{n^2 (n+1)}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$L = \lim_{n \rightarrow \infty} \left[\frac{(n+2)^2}{n^2 (n+1)} \right]$$

$$L = \lim_{n \rightarrow \infty} \left[\frac{n^2 + 4n + 4}{n^2 (n+1)} \right]$$

$$L = \lim_{n \rightarrow \infty} \left[\frac{1}{n+1} + \frac{4}{n^2} \right]$$

$$L = \frac{1}{\infty} + \frac{4}{\infty}$$

$$L = 0 + 0$$

$$L = 0 < 1$$

$\therefore$ By the limit form of the ratio test, the series $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{n^2 (n+1)^2}{n!}$ is absolutely & convergent.

$\therefore$ It is convergent.

$\therefore$ It is convergent.

④ Use the root test to determine the convergence of the following series.

① $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$

Let $a_n = \frac{1}{\left(1 + \frac{1}{n}\right)^{n^2}} > 0$ for $n \geq 1$

$\rightarrow |a_n| = \left| \frac{1}{\left(1 + \frac{1}{n}\right)^{n^2}} \right|$

$|a_n| = \frac{1}{\left(1 + \frac{1}{n}\right)^{n^2}}$ for $n \geq 1$

$\rightarrow |a_n|^{1/n} = \frac{1^{1/n}}{\left(1 + \frac{1}{n}\right)^{n^2 \times 1/n}}$

$|a_n|^{1/n} = \frac{1}{\left(1 + \frac{1}{n}\right)^n}$

$\alpha = \limsup |a_n|^{1/n}$

$\alpha = \limsup \frac{1}{\left(1 + \frac{1}{n}\right)^n}$

$\alpha = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n}$

Let $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = Y$

$\ln Y = \lim_{n \rightarrow \infty} \ln \left(1 + \frac{1}{n}\right)^n$

$\ln Y = \lim_{n \rightarrow \infty} n \ln \left(1 + \frac{1}{n}\right)$

$= \lim_{n \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{n}\right)}{\frac{1}{n}} \quad \frac{d/dn}{d/dn}$

$\ln Y = \lim_{n \rightarrow \infty} \frac{\left(\frac{-1/n^2}{1 + 1/n}\right)}{\left(\frac{-1/n^2}{1}\right)} \quad \frac{(-n^2)}{(-n^2)}$

$\ln Y = \lim_{n \rightarrow \infty} \frac{1}{1 + 1/n}$

$\ln Y = 1$

$Y = e$

$\alpha = 1/e$

$\alpha = 0.37 < 1$

$\therefore$ By the root test, the series is absolutely convergent.
 $\therefore$ It is convergent.

$$(ii) \sum_{n=1}^{\infty} \frac{n^{n^2}}{(n+1)^{n^2}}$$

In here $a_n = \frac{n^{n^2}}{(n+1)^{n^2}}$ for all $n \geq 1$

$$|a_n| = \left| \frac{n^{n^2}}{(n+1)^{n^2}} \right|$$

$$|a_n| = \left(\frac{n}{n+1} \right)^{n^2} \text{ for all } n \geq 1$$

$$\rightarrow |a_n|^{1/n} = \left(\frac{n}{n+1} \right)^{n^2 \times \frac{1}{n}}$$

$$|a_n|^{1/n} = \left(\frac{n}{n+1} \right)^n$$

$$\alpha = \limsup |a_n|^{1/n}$$

$$\alpha = \limsup \left[\left(\frac{n}{n+1} \right)^n \right]$$

$$\alpha = \lim_{n \rightarrow \infty} \left[\left(\frac{n}{n+1} \right)^n \right]$$

$$\alpha = \lim_{n \rightarrow \infty} \left[\frac{n/n}{n/n + 1/n} \right]^n$$

$$\alpha = \lim_{n \rightarrow \infty} \left[\frac{1}{1 + 1/n} \right]^n$$

$$\alpha = \lim_{n \rightarrow \infty} \frac{1}{(1 + 1/n)^n}$$

$$\text{Let } \lim_{n \rightarrow \infty} \left[1 + \frac{1}{n} \right]^n = y$$

$$\ln y = \ln \lim_{n \rightarrow \infty} \left[1 + \frac{1}{n} \right]^n$$

$$\ln y = \lim_{n \rightarrow \infty} n \ln \left[1 + \frac{1}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{\ln \left[1 + \frac{1}{n} \right]}{\frac{1}{n}} \quad \frac{d/dn}{d/dn}$$

$$\ln y = \lim_{n \rightarrow \infty} \left[\frac{\frac{-1/n^2}{1 + 1/n}}{-1/n^2} \right] \cdot \frac{(-n^2)}{(-n^2)}$$

$$\ln y = \lim_{n \rightarrow \infty} \frac{1}{1 + 1/n}$$

$$\ln y = 1$$

$$y = e$$

$$\alpha = \frac{1}{e}$$

$$\alpha = 0.37 < 1$$

$\therefore$ By the root test, the series is absolutely convergent.

$\therefore$ It is convergent.

$$\text{ii)} \sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \frac{x^{2n}}{2^n} = 2^n$$

$$\text{Let } a_n = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \frac{x^{2n}}{2^n} \quad \text{for all } n \in \mathbb{N}$$

$$|a_n| = \left| \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \frac{x^{2n}}{2^n} \right|$$

$$|a_n| = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \frac{x^{2n}}{2^n}$$

$$|a_n|^{1/n} = \left(\frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \frac{x^{2n}}{2^n} \right)^{1/n}$$

$$|a_n|^{1/n} = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{(2 \cdot 4 \cdot 6 \dots 2n)^{1/n}} \frac{x^2}{2^{1/n}}$$

$$|a_{n+1}| = \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{2 \cdot 4 \cdot 6 \dots (2n+2)} \frac{x^{2(n+1)}}{(2n+2)}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{2 \cdot 4 \cdot 6 \dots 2(n+1)} \frac{x^{2(n+1)}}{(2n+2)} \times \frac{2 \cdot 4 \cdot 6 \dots 2n}{1 \cdot 3 \cdot 5 \dots 2n} \frac{2^n}{x^{2n}} \right|$$

$$= \left| \frac{(2n+1) x^2}{2(n+1)} \right|$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(2n+1) x^2}{2(n+1)}$$

$$\alpha = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$\alpha = \lim_{n \rightarrow \infty} \left| \frac{(2n+1) x^2}{2(n+1)} \right|$$

$$\alpha = \frac{[2 + 1/n] x^2}{2(1 + 1/n)}$$

$$\alpha = \frac{2x^2}{2}$$

$$\alpha = x^2 \in \mathbb{R}$$

when $x \leq 1$

$\therefore$ The series $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \frac{x^{2n}}{2^n}$ is ~~divergent~~ absolutely convergent

when $x > 1$

$\therefore$ The series is divergent.

when $x = 1$

$\therefore$ The test fails

⑤ Use the integral test to determine the convergence or divergence of the following series.

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$$

Here, $a_n = \frac{1}{n(\log n)^p}$ for all $n \geq 2$

Let $f(x) = \frac{1}{x(\log x)^p}$ for all $x \geq 2$

i) $f(x) > 0$, $\forall x \in [2, \infty)$

ii) $f(x)$ is $\downarrow$

iii) $f(x)$ is continuous.

Now,

$$\lim_{n \rightarrow \infty} \int_2^n \frac{1}{x(\log x)^p} dx$$

$$\lim_{n \rightarrow \infty} \left[\frac{(\ln x)^{1-p}}{1-p} \right]_2^n$$

$$\lim_{n \rightarrow \infty} \left[\frac{(\ln n)^{1-p}}{1-p} - \frac{(\ln 2)^{1-p}}{1-p} \right]$$

$$\lim_{n \rightarrow \infty} \left[\frac{\ln(n)^{1-p} - (\ln 2)^{1-p}}{1-p} \right]$$

$$\lim_{n \rightarrow \infty} \left[\frac{(1-p) \ln n - (1-p) \ln 2}{(1-p)} \right]$$

$$\lim_{n \rightarrow \infty} \frac{(1-p)(\ln n - \ln 2)}{(1-p)}$$

$$\lim_{n \rightarrow \infty} \ln n - \ln 2$$

$$= \infty \notin (0, \infty)$$

discuss this with p value

$\therefore$ By the integral test, the series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ is divergent.

$$\textcircled{ii} \sum_{n=1}^{\infty} n e^{-n^2}$$

Here $a_n = n e^{-n^2}$ for all $n \geq 1$

Let $f(x) = x e^{-x^2}$ for all $x \geq 1$

- i) $f(x) > 0 \quad \forall x \in [1, \infty)$
- ii) $f(x) \downarrow$
- iii) $f(x)$ is continuous.

Now, $\lim_{n \rightarrow \infty} \left[\int_1^n x e^{-x^2} dx \right]$

$$\lim_{n \rightarrow \infty} \left[\int_1^n x \cdot e^u \cdot \frac{du}{-2x} \right]$$

$$\lim_{n \rightarrow \infty} \left[\int_1^n -\frac{1}{2} e^u du \right]$$

$$\lim_{n \rightarrow \infty} \left[-\frac{1}{2} \int_1^n e^u du \right]$$

$$\lim_{n \rightarrow \infty} \left[-\frac{1}{2} \{e^u\}_1^n \right]$$

$$\lim_{n \rightarrow \infty} \left[-\frac{1}{2} \{e^{-x^2}\}_1^n \right]$$

$$\lim_{n \rightarrow \infty} \left[-\frac{1}{2} e^{-n^2} - \frac{1}{2} e^{-1^2} \right]$$

$$\lim_{n \rightarrow \infty} \left[-\frac{1}{2} [e^{-n^2} - e^{-1}] \right]$$

$$-\frac{1}{2} \lim_{n \rightarrow \infty} [0 - 0.37]$$

$$0.185$$

$$u = -x^2$$

$$\frac{du}{dx} = -2x$$

$$dx = \frac{du}{-2x}$$

$$u=1,$$

$$-x^2=1$$

$$x= \pm 1$$

$$u=n,$$

$$-x^2=n,$$

$$x= \pm n.$$

$\therefore$ By the integral test, the series $\sum_{n=1}^{\infty} n e^{-n^2}$ is convergent

⑥ Determine the absolute convergence and convergence of the following series:

i) $\sum_{n=1}^{\infty} \frac{(-1)^n \cos n\alpha}{n\sqrt{n}}$

Let $a_n = \frac{(-1)^n \cos n\alpha}{n\sqrt{n}}$ for all $n \in \mathbb{N}$

$|a_n| = \frac{\cos n\alpha}{n\sqrt{n}}$

Then, $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{\cos n\alpha}{n\sqrt{n}} = \sum_{n=1}^{\infty} \frac{\cos n\alpha}{n^{3/2}}$

we know that $1/n^{3/2}$ series is absolutely convergent

$\left| \frac{\cos n\alpha}{n^{3/2}} \right| \leq \frac{1}{n^{3/2}}$

$\therefore$ By the comparison test, $\sum_{n=1}^{\infty} \frac{(-1)^n \cos n\alpha}{n^{3/2}}$ is also absolutely convergent.

$\therefore$ It is convergent.

ii) $\sum_{n=1}^{\infty} \frac{\cos n\alpha}{n^2}$

Let $a_n = \frac{\cos n\alpha}{n^2}$ for all $n \in \mathbb{N}$

$|a_n| = \left| \frac{\cos n\alpha}{n^2} \right|$

$\sum_{n=1}^{\infty} \cos n\alpha = \cos \alpha + \cos 2\alpha + \cos 3\alpha + \dots + \cos n\alpha$
 $= \frac{\cos n\alpha/2}{\cos \alpha/2} \times \cos \left[\alpha + (n-1) \frac{\alpha}{2} \right]$

$\left| \sum_{n=1}^{\infty} \cos n\alpha \right| = \left| \frac{\cos n\alpha/2}{\cos \alpha/2} \right| \left[\cos (n+1) \frac{\alpha}{2} \right]$

$\left| \sum_{n=1}^{\infty} \cos n\alpha \right| < \frac{1}{|\cos \alpha/2|} \times 1$

$\left| \sum_{n=1}^{\infty} \cos n\alpha \right| < \frac{1}{|\cos \alpha/2|}$

This is bounded.

we know that $\sum_{n=1}^{\infty} 1/n^2$ is convergent. $\therefore \sum_{n=1}^{\infty} \frac{\cos n\alpha}{n^2}$ is absolutely convergent

$$iv) \sum_{n=1}^{\infty} \frac{(-1)^n (n+2)}{2^n + 5} \cos nx$$

$$\text{Let } a_n = \frac{(-1)^n (n+2)}{2^n + 5} \cos nx \text{ for all } n \in \mathbb{N}$$

$$\text{So, } |a_n| = \frac{(n+2) \cos nx}{2^n + 5}$$

$$\text{Let } b_n = \frac{1}{n^2} \text{ for all } n \in \mathbb{N}$$

$$l = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

$$l = \lim_{n \rightarrow \infty} \left[\frac{\frac{(-1)^n (n+2)}{2^n + 5}}{1/n^2} \right]$$

$$l = \lim_{n \rightarrow \infty} \left[\frac{(-1)^n (n+2)}{2^n + 5} \times \frac{n^2}{1} \right]$$

$$l = \lim_{n \rightarrow \infty} \left[\frac{(-1)^n (n^3 + 2n^2)}{2^n + 5} \right]$$

$$l = \frac{n^3/n^3 + 2n^2/n^3}{2^n/n^3 + 5/n^3}$$

$$l = \frac{1}{0}$$

$l = \text{does not exist}$

$$\therefore \sum_{n=1}^{\infty} \frac{(-1)^n (n+2)}{2^n + 5} \text{ is divergent.}$$

$\therefore$ The series is not convergent.

(7) Suppose that $\sum_{n=1}^{\infty} a_n$ is a series of positive terms a_n which is convergent. Prove that $\sum_{n=1}^{\infty} a_n^2$ and $\sum_{n=1}^{\infty} \sqrt{a_n a_{n+1}}$ are convergent.

Since, $\sum_{n=1}^{\infty} a_n$ is convergent $\lim_{n \rightarrow \infty} a_n = 0$
 For all $\epsilon > 0$, there exists an integer N ,
 s.t. $|a_n| < \epsilon$ whenever $n \geq N$.

consider,

$$\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^N a_n^2 + \sum_{n=N+1}^{\infty} a_n^2$$

$\sum_{n=1}^N a_n^2$ is convergent because that series is a finite sum.

$\therefore \sum_{n=N+1}^{\infty} a_n^2$ is also convergent by the comparison test.

$\therefore \sum_{n=1}^{\infty} a_n^2$ is also convergent

Let consider,

$$\sum_{n=1}^{\infty} \sqrt{a_n a_{n+1}} \leq \sqrt{\sum_{n=1}^{\infty} a_n} \times \sqrt{\sum_{n=1}^{\infty} a_{n+1}}$$

$\sqrt{\sum_{n=1}^{\infty} a_n}$ is convergent because $\sum_{n=1}^{\infty} a_n$ is convergent

$\therefore \sqrt{\sum_{n=1}^{\infty} a_{n+1}}$ is also convergent by the comparison test.

$\therefore \sum_{n=1}^{\infty} \sqrt{a_n a_{n+1}}$ is convergent.

⑨ Use the comparison test to show that the series $\sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$ converges if $q > 1/2$,

$$\sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$$

In here,

$$a_n = \frac{1}{(n^2+n)^q}$$

Let $q = 1$,

$$\text{so, } a_n = \frac{1}{(n^2+n)}$$

$$\frac{1}{n^2+n} < \frac{1}{n^2} = b_n$$

$$\text{Let } a_n = \frac{1}{n^2+n} \text{ and } b_n = \frac{1}{n^2}$$

Then $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} 1/n^2$ is convergent.

$\therefore$ By the comparison test, $\sum_{n=1}^{\infty} a_n$ is convergent.

$\therefore \sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$ is convergent when $q > 1/2$

ii) Diverges if $q \leq 1/2$

$$a_n = \frac{1}{(n^2+n)^q}$$

Let $q = 1/2$

$$\text{so, } a_n = \frac{1}{\sqrt{n^2+n}}$$

$$\frac{1}{\sqrt{n^2+n}} < \frac{1}{\sqrt{n^2}} = \frac{1}{n} = b_n$$

$$\text{Let } a_n = \frac{1}{\sqrt{n^2+n}} \text{ and } b_n = \frac{1}{n}$$

Then $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} 1/n$ is divergent.

$\therefore$ By the comparison test, $\sum_{n=1}^{\infty} a_n$ is divergent

$\therefore \sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$ is divergent when $q \leq 1/2$